Name.....

Signature……..**Candidates Personal Number**…….

545/1

CHEMISTRY.

Paper 1.

Aug /2025.

2 hours

CHEMISTRY DEPARTMENT

New Lower Secondary Chemistry Examination

Uganda Certificate of Education

Internal mock set 8

Paper 1

2 hours

INSTRUCTIONS:

This paper consists of **two** *sections* **A** *and* **B**. *It has* **six** *examination items.*

Section **A** *has* **one compulsory** items.

Section **B** *has* **two parts, I** and **II**. *Answer* **one** item from each part.. Answer **four** items in all.

Answers to section **A must** *be written in the spaces provided while those of section* **B must** *be written on the provided answer sheets.*

Any additional item(s) answered will **not** *be scored*

Where necessary use, Na = 23, Cl = 35.5 ,*1 mole of gas occupies 22.4 liters at s.t.p.*

SECTION A

Answer **all** *items in this section in the spaces provided.*

Item 1

Mulago National referral Hospital has received a new radiopharmaceutical isotope X (Initial mass is 100 g) for cancer treatment. It forms nuclide **Y** and emits particle **P**, through a **decay process**, which can be used for other applications

$$^{223}_{88}\mathbf{X} \rightarrow ^{219}_{86}\mathbf{Y} + \mathbf{P}$$

Due to its short half life of **11** days, the team needs to use the sample for its intended purpose within specific timeframe. However, the patient's treatment was postponed and **34** days have passed. They need to determine the remaining mass of the sample to ensure it's still safe and effective and ensure safety procedures.

Task. As a chemistry student,

(a) identify **P**

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(b) calculate the mass of the remaining sample.

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(c) suggest any other use of the type of decay process that took place

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(d) explain the dangers associated with the applications of the type of decay process.

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Item 2.

A chemical and material manufacturing industry plans to utilize the properties of some elements of the periodic table below with high industrial value for their applications while ensuring their environmental consideration.

	I	II	III	IV	V	VI	VII	VIII
Period 2								
Period 3	**Na**		**Al**			**S**	**Cl**	
Period 4								

He needs to produce sodium chloride. A preliminary experiment will be carried out to know the volume of chlorine at s.t.p required to completely react with sodium to produce 117.6 g of sodium chloride. This will help to know the efficiency of the planned production line. You have been consulted you for guidance.

$$2Na\,(s) + Cl2(g) \text{ --------} \rightarrow 2NaCl\,(s)$$

Task: As a Chemistry student,

(a) explain the category of the substances in the table .

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(b) suggest the suitability of the group (III) element to its application in life.

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(c) Calculate the volume of chlorine required.

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(d) advise on the environmental impact associated with use of either group (III) or (VI) substance in the table.

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SECTION B

Part I (*Attempt one item from this section)*

Item 3 The vast majority of iron is used to produce steel an alloy of iron and carbon used for making bars of construction, tools and heavy machinery. To increase on quantities of iron on market needed , the government has cleared an investor to set up an iron extraction plant in kayunga . However ,the locals are concerned about its impact and are seeking assurance on how they will benefit from it's operations. The district officials have organized a workshop to educate the locals. You have been invited to attend.

Task.

As a chemistry student, make a write up you will use during the presentation.

Item 4.

Due to large quantities of sodium hydroxide needed on market for production of detergents and bleaching agents, Jinja Municipal council officials have cleared an investor to set up a sodium hydroxide manufacturing plant. However, the locals are concerned about its impact to the environment and how they will socially benefit from the plant. The officials have organized workshop at the municipal hall. You have been invited to address the locals.

Task.

Prepare a write up message you will use.

Part II

(*Attempt* **one** *item from part)*

Item 5

Uganda is actively pursuing the development of its first crude oil refinery to ensure domestic energy security. The refinery is planned to be built in kabaale village, Buseruka Sub county in Hoima district near the rich crude oil fields. However, residents are concerned about the potential for environmental damage from refinery process particularly on air. Official from ministry of energy have organized workshops to address the locals . You have been invited to address the locals

Task.

Write a write up presentation you will use during the presentation.

Item 6.

Many water sources in several districts are adversely affected by large scale crop cultivation and rapid establishment of medium -sized factories. While these endeavors have boosted the incomes of many families, they have become a source of significant concern for the district officials. In response, sensitization workshops have been organized to address the situation . You have been invited to address the public.

Task.

Prepare a write up message you will use.

End.